

MovementLogger — Sensor & Module Catalog

10 Teile · 2026-05-25 16:46 UTC · BOM aus movement_logger_firmware · via crawl2pump

10 neu (7 Tage)

0 aktualisiert

Dev Boards 3

SparkFun OpenLog Artemis (Apollo3 datalogger, microSD, USB-C, Qwiic)

OSS firmware

USB-C pluggable

☒ USB-C

☐ WiFi

☒ Bluetooth

☐ GPS

☒ Motion sensors

☒ SD-card

L×B×H 5.3 × 4.1 × 1.5 cm · 13 g

MCU: Apollo3 Blue · Cortex-M4F @48 MHz · 1 MB flash / 384 KB SRAM · BLE 5

Datasheet: <https://www.sparkfun.com/products/19426>

OSS firmware: https://github.com/sparkfun/OpenLog_Artemis



neu SparkFun OpenLog Artemis (Apollo3 datalogger, microSD, USB-C, Qwiic) · DEV-19426 @ DigiKey

CHF 38.13

SparkFun @ DigiKey · in stock

Standalone Qwiic datalogger. Apollo3 Blue Cortex-M4F @48 MHz + onboard ICM-20948 9-DoF IMU + MS5637 barometer + microSD slot + USB-C charge + BLE 5. Runs SparkFun's open OpenLog_Artemis firmware out of the box: power on, it logs every Qwiic device it detects (any GPS in this BOM) plus its own IMU and baro to a CSV/UBX on the SD card. Configure sample-rate, file naming, sleep behaviour via a USB serial menu. The natural pair for a small RTK GPS build: plug in a SparkFun ZED-F9P or MAX-M10S Qwiic breakout, hook up a LiPo, drop it in the SERPAC RBF33 IP67 case — a complete waterproof GPS data recorder, no firmware writing needed. · USB-C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/19426/16570916>

GPS / GNSS 15

Interface Cable SMA → U.FL, 100 mm (pigtail)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 10 × 0.3 × 0.3 cm · 3 g

Datasheet: <https://www.sparkfun.com/products/9145>



neu Interface Cable SMA → U.FL, 100 mm (pigtail) · SparkFun 9145 @ sparkfun.com

USD 5.75

SparkFun

100 mm pigtail: U.FL female (plugs into the SparkFun MAX-M10S breakout) ↔ SMA female bulkhead (mates with the antenna above's SMA male). Buy with the magnetic antenna or skip both if the onboard chip antenna is enough. · SMA / U.FL · firmware: open-source

<https://www.sparkfun.com/products/9145>

Molex Flexible GNSS Antenna u.FL (passive multi-band, paper-thin)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 4 × 1.54 × 0.01 cm · 2 g

Datasheet: <https://www.sparkfun.com/products/15246>



neu Molex Flexible GNSS Antenna u.FL (passive multi-band, paper-thin) · SparkFun 15246 @ sparkfun.com

USD 6.50

SparkFun

Passive flex GNSS-Patch (Molex 146236-0150 via SparkFun GPS-15246): 40 × 15 × 0.1 mm, adhesive backing, 50 mm u.FL pigtail — plugs straight onto the MAX-M10S / NEO-M9N Qwiic breakout's u.FL without the SMA pigtail (CAB-09145). Multi-band (L1 + L5 GPS, Galileo, BeiDou, GLONASS) so it stays useful if the receiver gets swapped to ZED-F9P. Passive (~+2 dBi gain) — better than the breakout's onboard chip antenna, worse than a +28 dB LNA active puck (PRT-14986). The smallest no-protrusion antenna option in the BOM: stick it flat to the inside of the RBF33's clear PC lid; the signal passes through the polycarbonate. Best for case-size-constrained builds where you'd rather accept a few dB of antenna loss than route an external puck + 3 m of coax. · SMA / U.FL · firmware: open-source

<https://www.sparkfun.com/products/15246>

SparkFun GPS Breakout - MAX-M10S (Qwiic)

OSS firmware

Qwiic / I²C

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

☐ Motion sensors

☐ SD-card

L×B×H 2.54 × 2.54 × 0.6 cm · 4 g

Datasheet: https://cdn.sparkfun.com/assets/7/5/9/a/a/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu SparkFun GPS Breakout - MAX-M10S (Qwiic) · GPS-18037 @ DigiKey

CHF 35.80

SparkFun @ DigiKey · in stock

Recommended MAX-M10S carrier. Open-hardware breakout. · Qwiic / I²C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/18037/16719314>

SparkFun GPS-RTK-SMA Breakout - ZED-F9P (Qwiic, multi-band RTK)

OSS firmware

Qwiic / I²C

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

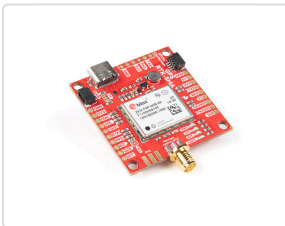
☐ Motion sensors

☐ SD-card

L×B×H 4.6 × 4.3 × 0.6 cm · 10 g

Datasheet: https://content.u-blox.com/sites/default/files/ZED-F9P-04B_DataSheet_UBX-21044850.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu SparkFun GPS-RTK-SMA Breakout - ZED-F9P (Qwiic, multi-band RTK) · GPS-16481 @ DigiKey

CHF 214.23

SparkFun @ DigiKey · in stock

DIY building block for a custom RTK tracker. u-blox ZED-F9P multi-band (L1+L2 GPS / Galileo / GLONASS / BeiDou) — same chip as the RTK Facet above, but as a bare Qwiic breakout for board-mount builds. Pair with any ESP32 dev board in this BOM (SparkFun Thing Plus-C, Seeed XIAO ESP32-S3) for WiFi + an NTRIP client; SparkFun ships the Arduino library `SparkFun u-blox GNSS Arduino Library` that wraps UBX protocol over I²C/Qwiic. SMA antenna jack onboard — wire to the u-blox ANN-MB-00 multi-band antenna below (L1+L2 active patch, 5 m coax, magnetic mount) for production performance. Without RTK corrections this is still a ~30 cm receiver under good sky; with RTK it's 1–2 cm. · Qwiic / I²C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/16481/12178437>

SparkFun NEO-M9N Chip Antenna Breakout (Qwiic, concurrent GNSS L1)

OSS firmware

Qwiic / I²C

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

☐ Motion sensors

☐ SD-card

L×B×H 2.6 × 1.7 × 0.6 cm · 5 g

Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_DataSheet_UBX-19014285.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu SparkFun NEO-M9N Chip Antenna Breakout (Qwiic, concurrent GNSS L1) · GPS-15733 @ DigiKey

CHF 59.59

SparkFun @ DigiKey · in stock

Concurrent four-GNSS receiver (GPS + Galileo + GLONASS + BeiDou, all L1) on a tiny 26 × 17 mm Qwiic breakout with onboard chip antenna. Position accuracy ~1.5 m standalone, ~0.6 m with SBAS corrections — the middle tier between the cheaper single-system MAX-M10S (slightly faster TTFF but ~2 m) and the multi-band ZED-F9P (RTK, cm-level). Update rate up to 25 Hz, ~30 mA. Plug into the OpenLog Artemis via Qwiic for a sub-meter-class standalone GPS logger without RTK complexity / subscription / NTRIP. · Qwiic / I²C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/15733/11481329>

u-blox ANN-MB-00 — multi-band active GNSS antenna (L1+L2+L5, SMA, 5 m)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 6 × 6 × 1.6 cm · 120 g

Datasheet: https://content.u-blox.com/sites/default/files/ANN-MB_DataSheet_UBX-18049862.pdf



neu u-blox ANN-MB-00 — multi-band active GNSS antenna (L1+L2+L5, SMA, 5 m) · ANN-MB-00 @ DigiKey

CHF 48.07

u-blox @ DigiKey · in stock

Multi-band L1/L2/L5 active patch antenna for the ZED-F9P. Magnetic base, 5 m RG-174 coax, SMA male plug — the antenna u-blox themselves spec for multi-band testing. Without a multi-band antenna the ZED-F9P falls back to L1-only and you lose the RTK precision advantage; pair these two. Mount the puck on a non-metal ground-plane (the foilboard deck or case lid) with clear sky view. · SMA / U.FL · firmware: open-source

<https://www.digikey.com/en/products/detail/u-blox/ANN-MB-00/9817928>

Battery / Power 3

SparkFun JST Jumper 2 Wire Assembly (JST-PH 2-pin extension)

OSS firmware

battery cell

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 15 × 0.5 × 0.3 cm · 2 g

Datasheet: <https://www.sparkfun.com/products/8670>



neu SparkFun JST Jumper 2 Wire Assembly (JST-PH 2-pin extension) · PRT-08670 @ DigiKey

CHF 1.00

SparkFun @ DigiKey · out of stock

JST-PH 2.0 mm 2-pin connector (female) → 2× bare stranded wires, ~150 mm. Use it to extend the LiPo lead inside the case (battery on one side, logger on the other), or to splice in a hard power-switch for a clean on/off without unplugging the cell. Polarity matches SparkFun LiPos — red on the same pin as the cell's red lead. Solder + heat-shrink to extend; for a JST-to-JST extension, twist two of these back to back. NOT a load-bearing power link on its own — too thin for sustained > 1 A draw, fine for the Artemis's ~50–80 mA logging current. · battery cell · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/08670/7393695>

SparkFun Lithium Ion Battery 1Ah, JST-PH 2-pin (OpenLog Artemis power)

OSS firmware

battery cell

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 6 × 3.6 × 0.7 cm · 22 g

Datasheet: <https://www.sparkfun.com/products/13813>



neu SparkFun Lithium Ion Battery 1Ah, JST-PH 2-pin
(OpenLog Artemis power) · SparkFun 13813 @ sparkfun.com

USD 16.95

SparkFun

1 Ah single-cell LiPo (3.7 V nominal) with JST-PH 2.0 mm 2-pin connector — plugs straight into the OpenLog Artemis (DEV-19426) battery jack. The Artemis's onboard MCP73831 charges it over USB-C (~500 mA) and runs the board untethered when unplugged. Runtime at 1 Hz GPS logging $\approx$ 12–20 h. Polarity is correct out of the box for SparkFun loads — verify with a multimeter before connecting to anything else. Mount with foam or velcro inside the case; an unrestrained LiPo under wave load can tear the JST leads off the cell. Class 9 lithium shipping rules apply: most distributors ship surface only, no air freight to private addresses. · battery cell · firmware: open-source

<https://www.sparkfun.com/products/13813>

Enclosure 1

SERPAC RBF33P06C10C — IP67 PC enclosure, clear lid (82 × 80 × 35 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 8.2 × 8 × 3.5 cm · 65 g



neu SERPAC RBF33P06C10C — IP67 PC enclosure, clear lid
(82 × 80 × 35 mm) · RBF33P06C10C @ SERPAC (vendor page)

USD 14.27

SERPAC (manufacturer store) · in stock

Polycarbonate IP67 project box, external 82 × 80 × 35 mm (3.23 × 3.15 × 1.38 in), clear PC lid sealed with perimeter O-ring + stainless-steel screws into metal inserts. Smallest RBF footprint that comfortably fits the OpenLog Artemis (53 × 41 mm) + a ZED-F9P Qwiic breakout (46 × 43 mm) stacked, plus a small LiPo. Same RF-transparent / Hall-magnet / Qi-charge tricks as the RBF53 / RBF63 line. For the GPS-only standalone logger build — STEVAL stays in its own RBF53 case nearby, GPS box sits separately on the deck with clear sky view. · enclosure · firmware: open-source · manufacturer page — price/stock also via Mouser/DigiKey/Farnell

<https://www.serpac.com/rbf33-electronic-enclosure.html>